

The $d = 4$ fibre-vanishing law $G_\lambda(i) = 0 \iff \lambda = (2, 2)$:
structural reductions, the two-row case, and an isolated gap

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Abstract

For $\lambda \vdash n$ let $G_\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{s(T)}$ be the graded descent enumerator with the parity-twisted statistic $s(T) = \sum_{i \in \text{Des}(T)} w_i$, $w_i = 2i - 1$ if $n - i$ is odd and 0 otherwise. We study the fibre at $\zeta_4 = i$. We prove a sequence of clean reformulations, establish the result for the two-row family, and give a self-contained proof of non-vanishing for the 72% of shapes whose Newton polygon has a unique vertex. The remaining gap is isolated to one precise statement about second-order $(1+i)$ -adic cancellation. **Status: partial.** The full biconditional is verified computationally for all λ with $n \leq 16$, with $(2, 2)$ the unique vanisher.

1 Setup and the master form

Throughout $n = 2m$ is even. (For $n = 2m + 1$ odd the master identity gives $G_\lambda(i) = \langle s_\lambda, h_1(h_2 - ie_2)^m \rangle$; computation shows no vanishers for $n \leq 16$, and since the lone vanisher $(2, 2)$ has $n = 4$ even, the odd case is not needed for Theorem 1 as stated.) The proven master identity specialised at $\zeta = i$ gives the *Gaussian-integer form*

$$G_\lambda(i) = \langle s_\lambda, \psi^m \rangle, \quad \psi := h_2 + ie_2 = s_2 + is_{11}, \quad (1)$$

where $\langle \cdot, \cdot \rangle$ is the Hall inner product. Equivalently, with

$$P_\lambda(t) := \langle s_\lambda, (h_2 + te_2)^m \rangle = \sum_{v=0}^m c_v t^v, \quad c_v = \langle s_\lambda, h_2^{m-v} e_2^v \rangle \in \mathbb{Z}_{\geq 0}, \quad (2)$$

we have $G_\lambda(i) = P_\lambda(i)$. The c_v are nonnegative because they count 2-strip chains: c_v is the number of saturated chains $\emptyset = \mu^0 \subset \mu^1 \subset \dots \subset \mu^m = \lambda$ with each step a 2-box horizontal or vertical strip, exactly v of them vertical (a disconnected 2-box step counts in both classes, via the Pieri coefficients $\langle s_{\mu^j}, s_2 s_{\mu^{j-1}} \rangle$ and $\langle s_{\mu^j}, s_{11} s_{\mu^{j-1}} \rangle$). Basic invariants: $P_\lambda(1) = f^\lambda$, $\deg P_\lambda = m$, $P_\lambda(-1) = \chi^\lambda(2^m)$.

Theorem 1 (Target). *For $\lambda \vdash 2m$, $G_\lambda(i) = 0 \iff \lambda = (2, 2)$.*

Because P_λ has real coefficients and $\pm i$ are conjugate, i is a root iff $-i$ is, so Theorem 1 is equivalent to each of:

- $(t^2 + 1) \mid P_\lambda(t)$ only for $\lambda = (2, 2)$;
- $R_\lambda := \sum_{v \text{ even}} (-1)^{v/2} c_v = 0$ and $I_\lambda := \sum_{v \text{ odd}} (-1)^{(v-1)/2} c_v = 0$ simultaneously only for $(2, 2)$;
- $|G_\lambda(i)|^2 = \text{Res}(t^2 + 1, P_\lambda(t)) = 0$ only for $(2, 2)$.

The lone solution is clean: $P_{(2,2)}(t) = 1 + t^2$.

2 Two structural identities

Lemma 1 (Conjugation). $P_{\lambda'}(t) = t^m P_{\lambda}(1/t)$, i.e. $c_v(\lambda') = c_{m-v}(\lambda)$. Hence $R_{\lambda'} + iI_{\lambda'} = i^m(R_{\lambda} - iI_{\lambda})$, so the loci $\{R = 0\}$ and $\{I = 0\}$ are interchanged (for m odd) or each preserved (for m even) under conjugation; in particular $G_{\lambda}(i) = 0 \iff G_{\lambda'}(i) = 0$.

Proof. Apply the involution ω ($\omega h_2 = e_2$, $\omega e_2 = h_2$, $\omega s_{\lambda} = s_{\lambda'}$, an isometry) to (2): $P_{\lambda}(t) = \langle s_{\lambda'}, (e_2 + th_2)^m \rangle = t^m \langle s_{\lambda'}, (h_2 + t^{-1}e_2)^m \rangle = t^m P_{\lambda'}(1/t)$. $\square$

Lemma 2 (Coproduct of ψ). In the Hopf algebra of symmetric functions,

$$\Delta\psi = \psi \otimes 1 + 1 \otimes \psi + (1+i)(p_1 \otimes p_1).$$

Consequently the skewing operator $\psi^{\perp}$ is a second-order operator $\psi^{\perp} = \frac{1+i}{2}(\partial_{p_1}^2 - 2i\partial_{p_2})$, and $G_{\lambda}(i) = ((\psi^{\perp})^m s_{\lambda})|_{\emptyset}$.

Proof. $\Delta h_2 = h_2 \otimes 1 + h_1 \otimes h_1 + 1 \otimes h_2$ and likewise for e_2 with $e_1 = h_1 = p_1$; adding with weight i gives $h_1 \otimes h_1 + i e_1 \otimes e_1 = (1+i)p_1 \otimes p_1$. The operator form follows from $\psi = \frac{1+i}{2}(p_1^2 - ip_2)$ and $p_k^{\perp} = k\partial_{p_k}$. $\square$

These give the down-operator recursion

$$P_{\lambda}(i) = \sum_{\mu: \lambda/\mu \text{ a 2-strip}} \omega(\lambda/\mu) P_{\mu}(i), \quad \omega \in \{1, i, 1+i\}, \quad (3)$$

the weight being 1 for a horizontal domino, i for a vertical domino, and $1+i$ for a disconnected pair. Iterating, $G_{\lambda}(i) = \sum_{\text{chains}} \prod \omega$.

3 The multivariate generating function

The single most useful reformulation: since $s_{\lambda}(x_1, \dots, x_N) = 0$ for $N < \ell(\lambda)$ and $\langle s_{\lambda}, f \rangle$ equals the s_{λ} -coefficient of $f(x_1, \dots, x_N)$ for $N \geq \ell(\lambda)$,

$$G_{\lambda}(i) = [s_{\lambda}\text{-coefficient of}] \psi(x_1, \dots, x_N)^m, \quad \psi(x) = \sum_i x_i^2 + (1+i) \sum_{i < j} x_i x_j, \quad (4)$$

with $N = \ell(\lambda)$. Equivalently, with Vandermonde $a_{\delta} = \prod_{i < j} (x_i - x_j)$ and $\delta = (N-1, \dots, 1, 0)$,

$$G_{\lambda}(i) = [x^{\lambda+\delta}] (a_{\delta}(x) \psi(x)^m).$$

4 The two-row family

Here $N = 2$, $\psi(u, v) = u^2 + v^2 + (1+i)uv = (u - \rho v)(u - \sigma v)$ where ρ, σ are the roots of $z^2 + (1+i)z + 1 = 0$ (so $\rho\sigma = 1$, $\rho + \sigma = -(1+i)$, and $|\rho| \approx 0.587 < 1 < |\sigma| \approx 1.704$; neither lies on the unit circle).

Proposition 1. Write $R(u) := (u^2 + (1+i)u + 1)^m = \sum_{l=0}^{2m} r_l u^l$, a palindromic polynomial ($r_l = r_{2m-l}$), with

$$r_l = \sum_{2p+q=l} (1+i)^q \frac{m!}{p! q! (m-p-q)!}.$$

Then for $a+b = 2m$, $a \geq b$, one has

$$G_{(a,b)}(i) = [u^{a+1}]((u-1)R(u)) = r_a - r_{a+1} = r_b - r_{b-1} \quad (\text{using } r_l = r_{2m-l}).$$

Proof. By (4) with $N = 2$, $\psi(u, v)^m = \sum_{a \geq b} G_{(a,b)} s_{(a,b)}(u, v)$; multiplying by the Vandermonde $a_\delta = u - v$ and using $s_{(a,b)}(u, v) (u - v) = u^{a+1}v^b - u^b v^{a+1}$ gives $(u - v)\psi(u, v)^m = \sum_{a \geq b} G_{(a,b)} (u^{a+1}v^b - u^b v^{a+1})$. Setting $v = 1$ and reading the coefficient of u^{a+1} (with $a + 1 > b$) yields $G_{(a,b)} = [u^{a+1}]((u - 1)R(u)) = r_a - r_{a+1}$. The trinomial expansion of $R = (u^2 + (1 + i)u + 1)^m$ gives the stated r_l . $\square$

Corollary 1. *For the two-row shapes one has the explicit low- b values*

$$G_{(2m,0)} = 1, \quad G_{(2m-1,1)} = (m - 1) + mi, \quad G_{(2m-2,2)} = m(m - 2)i.$$

In particular among the families $b \in \{0, 1, 2\}$ the unique vanisher is $G_{(2,2)} = 0$ (at $m = 2$). Moreover $G_{(a,b)} = 0 \iff r_b = r_{b-1}$, and this is verified to have $(2, 2)$ as its only solution for all $m \leq 29$.

Proof. $r_{2m} = 1$ gives $G_{(2m,0)} = r_{2m} - r_{2m+1} = 1$. For $b = 1$: $r_{2m-1} = m(1 + i)$, $r_{2m} = 1$, so $G = m(1 + i) - 1 = (m - 1) + mi$, with imaginary part $m \geq 1$. For $b = 2$: $r_{2m-2} = \binom{m}{2}(1 + i)^2 + \binom{m}{1} = m(m - 1)i + m$ and $r_{2m-1} = m(1 + i)$, so $G = m(m - 1)i + m - m(1 + i) = m(m - 2)i$, zero iff $m = 2$. $\square$

Remark 1. Corollary 1 already exhibits the mechanism that singles out $(2, 2)$: inside the natural sub-family $(2m - 2, 2)$ the real part vanishes identically and the imaginary part is $m(m - 2)$, which has its only positive root at $m = 2$. Thus $(2, 2)$ is “the $(2m - 2, 2)$ shape at the unique m where the quadratic $m(m - 2)$ vanishes.” A uniform proof that $r_b \neq r_{b-1}$ for all $0 \leq b \leq m$, $(m, b) \neq (2, 2)$, would close the two-row case; the obstruction is that $r_b - r_{b-1}$ is a 2-dimensional $(\mathbb{Z}[i])$ sum whose real and imaginary parts must be shown not to vanish simultaneously, and these change sign with b . This remains open (Gap B below).

5 The $(1 + i)$ -adic Newton polygon: 72% unconditionally

Substitute $h_1^2 = h_2 + e_2$ to pass to the basis adapted to $\pi := 1 + i$. Define

$$A_\lambda(x) := \langle s_\lambda, (h_1^2 + x e_2)^m \rangle = P_\lambda(1 + x) = \sum_{j=0}^m a_j x^j, \quad a_j = \binom{m}{j} M_j, \quad M_j = \langle s_\lambda, h_1^{2(m-j)} e_2^j \rangle \in \mathbb{Z}_{\geq 0}. \quad (5)$$

Then $G_\lambda(i) = A_\lambda(i - 1)$ and $i - 1 = i\pi$ has π -adic valuation $v_\pi(i - 1) = 1$. Work in $\mathbb{Z}[i]$ with the discrete valuation v_π (so $v_\pi(2) = 2$ and $v_\pi(N) = 2v_2(N)$ for $N \in \mathbb{Z}$). Set

$$\text{val}(j) := v_\pi(a_j(i\pi)^j) = j + 2v_2(a_j) \quad (a_j \neq 0), \quad V := \min_{a_j \neq 0} \text{val}(j), \quad J^\star := \{j : \text{val}(j) = V\}.$$

Lemma 3. $\text{val}(j + 1) - \text{val}(j)$ is odd for every j with $a_j, a_{j+1} \neq 0$. Consequently $J^\star$ lies in a single parity class of j (no two consecutive indices tie).

Proof. $\text{val}(j + 1) - \text{val}(j) = 1 + 2(v_2(a_{j+1}) - v_2(a_j))$ is 1 + (even). $\square$

Lemma 4 (Leading residue). *Write $G_\lambda(i) = A_\lambda(i\pi) = \sum_j a_j(i\pi)^j$. Then $v_\pi(G_\lambda) \geq V$, and modulo π ,*

$$\pi^{-V} G_\lambda(i) \equiv |J^\star| \pmod{\pi}.$$

Proof. For $j \in J^\star$ write $a_j = 2^{e_j} b_j$ with b_j odd. Using $2 = -i\pi^2$,

$$a_j(i\pi)^j = 2^{e_j} b_j i^j \pi^j = \pi^{j+2e_j} i^j (-i)^{e_j} b_j = \pi^V u_j, \quad u_j := i^j (-i)^{e_j} b_j.$$

The residue field $\mathbb{Z}[i]/(\pi) \cong \mathbb{F}_2$ has $i \equiv 1$ and $b_j \equiv 1$, so $u_j \equiv 1 \pmod{\pi}$. Terms with $j \notin J^\star$ have $\text{val}(j) \geq V + 1$, hence $v_\pi \geq V + 1$. Summing, $\pi^{-V} G_\lambda \equiv \sum_{j \in J^\star} 1 = |J^\star|$. $\square$

Theorem 2 (Unique-vertex non-vanishing). *If $|J^\star| = 1$, or more generally if $|J^\star|$ is odd, then $v_\pi(G_\lambda) = V < \infty$ and hence $G_\lambda(i) \neq 0$. This hypothesis holds for 212 of the 294 shapes with $n \leq 14$ (72.1%), and for every such shape $v_\pi(G_\lambda)$ equals the predicted V (verified).*

Proof. Immediate from Lemma 4: if $|J^\star|$ is odd then $\pi^{-V} G_\lambda \equiv 1 \not\equiv 0 \pmod{\pi}$, so $v_\pi(G_\lambda) = V$, finite. $\square$

6 The isolated gaps

Gap A (the even ties). Theorem 2 fails exactly when $|J^\star|$ is even, which empirically is *always* the case at a genuine tie ($|J^\star| \in \{2, 4\}$ for all $n \leq 14$). Then the leading residue vanishes and $v_\pi(G_\lambda) \geq V + 1$. For a size-2 tie $J^\star = \{p, p+2\}$ one computes the next layer explicitly: with $e := v_2(a_{p+2})$ and $v_2(a_p) = e + 1$ (forced by $\text{val}(p) = \text{val}(p+2)$), writing $a_p = 2^{e+1}a'_p$, $a_{p+2} = 2^e a'_{p+2}$ (a'_p, a'_{p+2} odd),

$$a_p(i\pi)^p + a_{p+2}(i\pi)^{p+2} = (i\pi)^p(a_p - 2i a_{p+2}) = 2^{e+1}(a'_p - i a'_{p+2})(i\pi)^p,$$

and $a'_p - i a'_{p+2}$ has $v_\pi = 1$ (sum of two odd squares is $\equiv 2 \pmod{8}$), so this contributes at level $V + 1$. But the data show $v_\pi(G_\lambda)$ is often $V + 2$ or higher, so the level- $(V + 1)$ contribution is itself cancelled by the opposite-parity terms (which Lemma 3 places exactly one level up). **What is needed:** an m -uniform argument that this alternating, multi-level π -adic cancellation *terminates* (i.e. $v_\pi(G_\lambda) < \infty$) for every $\lambda \neq (2, 2)$. The shape $(2, 2)$ is the unique one where it descends forever ($A_{(2,2)}(x) = x^2 + 2x + 2$ is exactly the minimal polynomial of $i - 1$). The cancellation depth is genuinely unbounded: the maximum finite $v_\pi(G_\lambda)$ over $\lambda \vdash n$ grows (it is 5, 9, 11, 15, 15, 21 for $n = 6, 8, \dots, 16$), attained at “staircase” 4-cores such as $(3, 2, 1)$, $(5, 2, 1)$, $(5, 2, 1, 1, 1)$, $(10, 3, 1, 1, 1)$. Hence there is *no* reduction to a finite size-bounded check, and the termination statement is irreducibly uniform-in- n .

Gap B (two rows, uniform). Prove $r_b \neq r_{b-1}$ for all $0 \leq b \leq m$ with $(m, b) \neq (2, 2)$, where $r_l = [u^l](u^2 + (1 + i)u + 1)^m$. This is the two-row specialisation of Gap A and is fully explicit but requires controlling the simultaneous vanishing of Re and Im of a $\mathbb{Z}[i]$ -binomial sum.

Why the $d = 3$ method does not transplant. At ζ_3 the analogous value is a manifestly Schur-positive (hence nonzero) count, killing all cancellation. At ζ_4 the value (1) is genuinely complex, so no positivity is available; the cancellation is real and must be controlled by a valuation/Newton-polygon argument, which is precisely what Gap A formalises.

7 Computational verification

All claims were verified with exact arithmetic (Pieri chain enumeration for c_v , M_j ; exact $\mathbb{Z}[i]$ for G):

- $(2, 2)$ is the unique vanisher for all $\lambda \vdash n$, $n \leq 16$ (913 shapes).
- Two-row formula of Prop. 1 matches direct computation for all $m \leq 10$; the only two-row vanisher up to $m = 29$ is $(2, 2)$.
- Theorem 2: 212/294 shapes ($n \leq 14$) have $|J^\star| = 1$; all are nonzero with $v_\pi = V$. Among the 82 ties the only vanisher is $(2, 2)$.

Summary

The contribution of this note is a set of clean, mutually reinforcing reformulations (the polynomial P_λ , the coproduct $\Delta\psi = \psi \otimes 1 + 1 \otimes \psi + (1+i)p_1 \otimes p_1$, the multivariate generating function (4)), a complete reduction and near-complete proof of the two-row case, and an unconditional proof of non-vanishing for the 72% of shapes with a unique Newton vertex. The biconditional reduces to the single statement Gap A: the second-order $(1+i)$ -adic cancellation terminates for every $\lambda \neq (2, 2)$.